

01. Basic Programs — 1-16

1. Hello World
2. Print Name
3. Print College Details
4. Print Multiple Lines
5. Print ASCII Values
6. Add Two Numbers
7. Subtract Two Numbers
8. Multiply Two Numbers
9. Divide Two Numbers
10. Find Remainder
11. Swap Two Numbers (Using Third Variable)
12. Swap Without Third Variable
13. Square of a Number
14. Cube of a Number
15. Area of Circle
16. Area of Rectangle

02. Decision Making — 17-36

17. Calculator Using if-else
18. BMI Calculator
19. Electricity Bill Calculator
20. Income Tax Calculator
21. Even or Odd
22. Positive, Negative or Zero
23. Largest of Two Numbers

- 24.Largest of Three Numbers
- 25.Smallest of Three Numbers
- 26.Leap Year Check
- 27.Vowel or Consonant
- 28.Alphabet, Digit or Special Character
- 29.Uppercase or Lowercase
- 30.Grade Calculator
- 31.Divisibility Check
- 32.Percentage Calculator
- 33.Profit or Loss Calculator
- 34.Age Category
- 35.Voting Eligibility
- 36.Calculator Using match-case

03. Loop Programs — 37-50

- 37.Print N to 1
- 38.Sum of N Numbers
- 39.Sum of Even Numbers
- 40.Sum of Odd Numbers
- 41.Prime Numbers in Range
- 42.Reverse Number
- 43.Palindrome Number
- 44.Multiplication Table
- 45.Reverse Multiplication Table
- 46.Factorial
- 47.Fibonacci Series

48.Prime Number

49.Print 1 to N

50.Number Guessing Game

04. String Programs — 51-61

51.Find String Length

52.Reverse String

53.Palindrome String

54.Count Vowels

55.Count Consonants

56.Count Digits

57.Count Spaces

58.Count Words

59.Count Characters

60.Character Frequency

61.Remove Spaces

05. List and Tuple Programs — 62-82

62.Create List

63.Sum of List

64.Average

65.Maximum Element

66.Minimum Element

67.Second Largest

68.Second Smaller

69.Reverse List

70.Sort List

- 71. Search Element
- 72. Even Elements
- 73. Create Tuple
- 74. Convert List to Tuple
- 75. Tuple Packing
- 76. Tuple Unpacking
- 77. Nested Tuple
- 78. Count Elements
- 79. Maximum Element
- 80. Minimum Element
- 81. Tuple Slicing
- 82. Tuple Sorting

06. Set Programs — 83-90

- 83. Union
- 84. Intersection
- 85. Subset Check
- 86. Superset Check
- 87. Difference
- 88. Symmetric Difference
- 89. Set Difference Update
- 90. Set Length and Element Check

07. Functions — 91-99

- 91. Simple Function
- 92. Function with Arguments
- 93. Armstrong Number Using Function

- 94.Fibonacci Using Function
- 95.Factorial Using Recursion
- 96.Find Maximum Using Function
- 97.GCD Using Function
- 98.Palindrome Using Function
- 99.Prime Number Using Function

08. Searching and Functions — 100-109

- 100. Palindrome Using Function
- 101. GCD Using Function
- 102. Fibonacci Using Function
- 103. Factorial Using Recursion
- 104. Linear Search
- 105. Binary Search
- 106. Bubble Sort
- 107. Insertion Sort
- 108. Factorial Using Function
- 109. Selection Sort

09. Pattern Programs — 110-116

- 110. Rectangle
- 111. Right Triangle
- 112. Left Triangle
- 113. Inverted Triangle
- 114. Pyramid
- 115. Inverted Pyramid
- 116. Square

10. Mini Projects — 117-120

- 117. *Calculator Application*
- 118. *Student Grade Management System*
- 119. *Quiz Application*
- 120. *Number Guessing Game*